

The Basic Strong/Weak Problems

Opt. - last lecture

Violation

Validity

Separation

Membership.

Convex set

$$K \subseteq \mathbb{R}^n,$$

Violation

Strong Violation: SVIOL

Given $c \in \mathbb{R}^n$, $\gamma \in \mathbb{R}$ decide if $\exists x \leq \gamma, \forall x \in K$
and if not, find $y \in K$ with $c^T y > \gamma$.

Weak Violation: WVIOL

Given $c \in \mathbb{R}^n$, $\gamma \in \mathbb{Q}$, $\varepsilon \in \mathbb{Q} > 0$, either

- (i) Assert $c^T x \leq \gamma + \varepsilon, \forall x \in S(K, -\varepsilon)$
- (ii) Find $y \in S(K, \varepsilon)$ with $c^T y \geq \gamma - \varepsilon$

Validity

Strong Validity: SVAL

Given $c \in \mathbb{R}^n$, $\gamma \in \mathbb{R}$, decide if

$$c^T x \leq \gamma \quad \forall x \in K$$

Weak Validity: WVAL

Given $c \in \mathbb{Q}^n$, $\gamma \in \mathbb{Q}$, $\varepsilon \in \mathbb{Q}_{>0}$, either

- (i) Assert $c^T x \leq \gamma + \varepsilon \quad \forall x \in S(k, -\varepsilon)$ or
(ii) Assert $c^T y \geq \gamma - \varepsilon$ for some $y \in S(k, \varepsilon)$

Separation

Strong Separation: SSEP

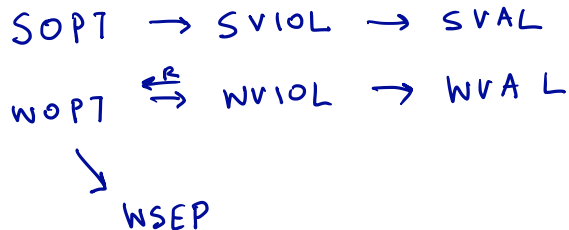
Given $y \in \mathbb{R}^n$, decide if $y \in K$ and if not,
find $c \in \mathbb{R}^n$ st $c^T y > \max \{c^T x : x \in K\}$
i.e., $c^T y > c^T x, \forall x \in K.$

Weak Separation: WSEP

Given $y \in \mathbb{R}^n$, $\delta \in \mathbb{Q}_{>0}$, either

- (i) Assert $y \in S(K, \delta)$ or
- (ii) Find $c \in \mathbb{Q}^n$, with $\|c\|_\infty = 1$, s.t. $c^T y \geq c^T x - \delta \quad \forall x \in S(K, -\delta)$

Relative Strength of Oracles



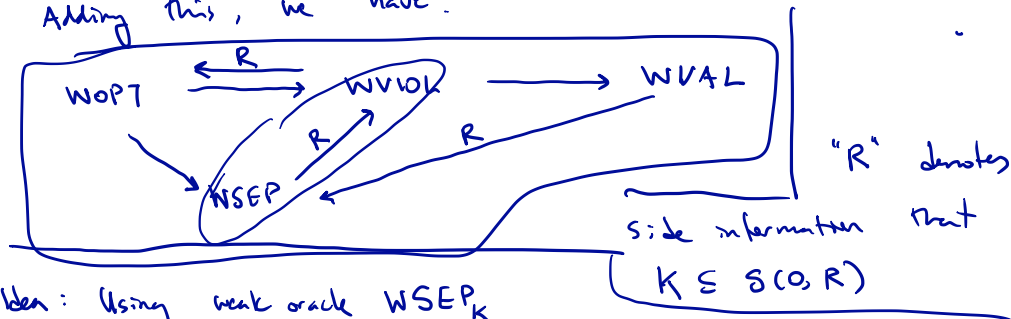
$$\text{WVIOL} \xrightarrow{R} \text{WOPT} \quad \text{by bisection}$$

We need a circumscribing ball.

Power of the Ellipsoid Algorithm

$WSEP \xrightarrow{R} WVIOK$ by Ellipsoid.

Adding this, we have:



Idea: Using weak oracle $WSEP_K$ make a weak separation oracle for run Ellipsoid algorithm with this separation oracle for $\hat{K}$.

$$\hat{K} = \underline{\underline{K \cap \{x : c^T x \geq \delta\}}}$$

Duality/Polarity of SEP and OPT

Defn: $P = \{x: Ax \leq b\}$, the

polar of P is defined as

$$P^* = \{y: x^T y \leq 1, \forall x \in P\}$$

Ex: $P = [-1, 1]^n$, $P^* = \{x: \|x\|_1 \leq 1\}$ Cross Polytope.

Thm: Solving an LP over P is equivalent to solving the separation problem over P^* .

pf: $WOPT \Rightarrow WRIO$ by bisection. Now show
 $WRIO$ linked via duality/polar to $WS \& P$.

Given P , $c \in \mathbb{R}^n$, consider decision problem:

$$\begin{cases} \max: c^T x \\ \text{st: } x \in P \end{cases} \leq 1$$

if not produce
 $y \in P^*$ st $c^T y > 1$.

Now: " $\max\{c^T x: x \in P\} \leq 1$ " $\Leftrightarrow$ " $c \in P^*$ "

Moreover, if $\exists y \in P$ wr $c^T y > 1 \Rightarrow c \notin P^*$ & $y^T c > y^T z, \forall z \in P^*$.